

QUE · INVESTOR / SE TECH DILIGENCE

# Technical Diligence, Production Sell Gate & Build-to-Compete Roadmap

Product: **Que** (SchemaGraph) · Codebase: `adc/schemagraph`

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Audience: founders, technical investors, sales engineers · Scope: **technology only** (not GTM/pricing)

**Verdict:** Conditional pass for a *scoped paid technical POC*. **Not** for “replace Atlan/Collibra this quarter.”  
**Not** for unattended multi-tenant SaaS at Big-4 audit bar — yet. Wedge that holds: *join truth* → *human approve* → *freeze* → *attested PR* on schema metadata only.

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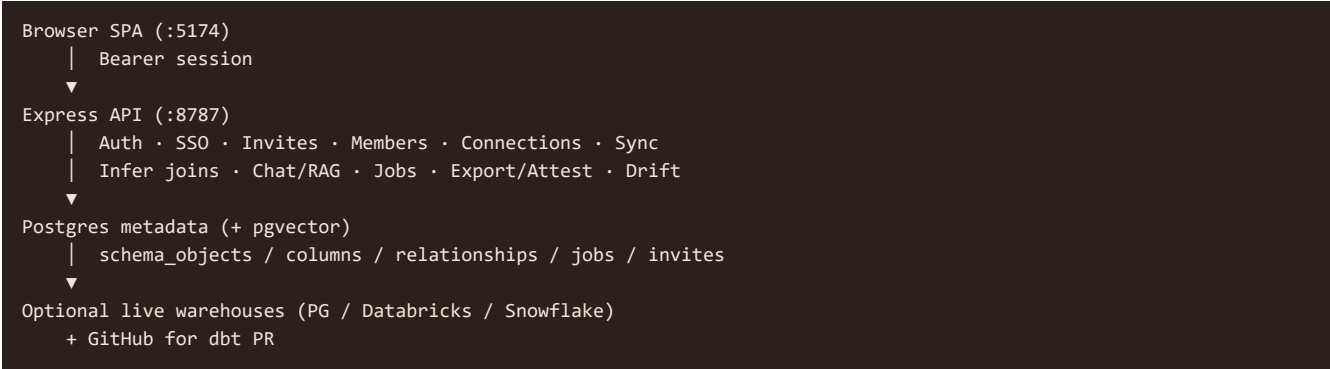
# 1. Executive tech verdict

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Que is a coherent **data-engineering workflow appliance** for cross-source join discovery, human-in-the-loop approval, contract freeze, and HMAC-attested export / dbt PR — on a **schema-metadata plane** (not a warehouse row lake).

| Question                                           | Answer                                                                                  |
|----------------------------------------------------|-----------------------------------------------------------------------------------------|
| Can we sell a controlled paid POC?                 | <b>Yes</b> — with fixed SOW (2–3 sources, success = promoted joins + attested PR).      |
| Can we claim “replace Atlan/Colibra this quarter”? | <b>No</b> — missing glossary depth, stewardship, lineage breadth, enterprise packaging. |
| Unattended multi-tenant SaaS at Big-4 bar?         | <b>No</b> — missing SOC2 evidence, SCIM, HA/isolation proof, pen-test/SLA posture.      |
| Is the wedge technically real?                     | <b>Yes</b> — HITL joins + freeze + attestation is implemented and testable.             |

## 2. What Que is (architecture)



| Layer          | Implementation today                                                              |
|----------------|-----------------------------------------------------------------------------------|
| Metadata plane | Postgres; encrypted connection secrets ( __enc ); workspace settings              |
| Connectors     | PostgreSQL, Databricks, Snowflake, Excel/CSV, Mongo — fixture and/or live         |
| Graph / HITL   | Canvas; suggest joins; Promote/Reject; stitch session                             |
| AI             | Schema packs, skills, optional LLM/RAG; sample previews from sync samples         |
| Ship path      | Notebook jobs → dry-run / live validate → JSON/SQL/dbt + HMAC → optional GH PR    |
| Security spine | Register/login, roles, invites, OIDC PKCE+JWKS, CORS/prod boot gates, scrub/drift |

**Frame:** join-HITL appliance — not catalog of record, not Unity/Horizon replacement, not Collibra governance OS.

## 3. Market gaps (practitioner complaints)

Patterns from PeerSpot / support threads / practitioner discourse on Atlan, Alation, Collibra, DataHub, and Unity Catalog ↔ Horizon/Iceberg debates:

| Pain                                                             | Who feels it                                   | Implication for Que                                               |
|------------------------------------------------------------------|------------------------------------------------|-------------------------------------------------------------------|
| Column-level / cross-tool lineage incomplete or expensive to run | DataHub, Alation (+ often Manta), all catalogs | Users want <i>usable</i> join truth, not another incomplete graph |
| Heavy setup / connector complexity                               | DataHub OSS especially                         | POC time-to-value is a weapon — keep fixtures + roadmap           |
| Governance UX for stewards, not DEs shipping PRs                 | Collibra, Alation, partly Atlan                | “PRs not portals” is the wedge — stay there                       |
| Catalog ≠ action (still hand-write dbt joins)                    | All catalogs                                   | Attested export/PR is the differentiator <i>if trustworthy</i>    |
| Cross-cloud SF↔DBX catalog politics                              | Unity vs Horizon                               | Cross-source help neither warehouse owns                          |
| AI suggestions lack trust without HITL                           | All AI catalogs                                | Promote/Reject is correct; never auto-accept joins                |

**What people say is missing:** not another search box — **trusted, reviewable, shippable join contracts across systems** with privacy-safe defaults.

## 4. Competitive tech matrix

| Capability                                  | Atlas / Alation / Collibra / DataHub | Unity / Horizon             | Que today                                       |
|---------------------------------------------|--------------------------------------|-----------------------------|-------------------------------------------------|
| Asset breadth / glossary / stewardship      | Strong                               | Platform-native             | Weak (by design)                                |
| Automated multi-hop column lineage at scale | Stronger when configured             | Strong in-platform          | Weak-partial (inference + optional DBX history) |
| Cross-source join HITL → freeze             | Rare as first-class loop             | In-platform only            | <b>Core — real</b>                              |
| Attested schema-only export / PR            | Rare                                 | N/A as product              | <b>Core — real</b>                              |
| Time-to-value for DE POC                    | Weeks-months                         | Fast if already on platform | <b>Days</b> with fixtures/docs                  |
| Enterprise IdP (SCIM), SOC2 story           | Expected                             | Expected                    | <b>Not there</b>                                |
| Query / warehouse governance                | Partial                              | Native                      | Out of scope                                    |

You do **not** beat them on catalog completeness. You can beat them on a **narrow DE workflow they under-serve**.

## 5. What Que has built (current capability)

### Access & tenancy

- Create account / sign in; optional OIDC SSO (PKCE, JWKS, hash token)
- Create / switch workspaces; invites + live members; role change / remove
- Roles: viewer → member → admin → owner
- In-app roadmap dialog (? header) with Skip / Next / Done

### Sources & schema

- Connectors: Postgres, Databricks, Snowflake, Excel/CSV, Mongo
- Encrypted credentials at rest; sync → canvas; join inference; Promote/Reject; stitch
- Optional Databricks query-history join assist

### Privacy & AI

- Schema-only defaults; sample scrub; policy toggles; BYOK LLM/GitHub secrets
- Chat skills + optional LLM/RAG; capped sample previews; job draft → Save → navigate to Jobs

### Jobs & ship

- Notebook cells; dry-run; live validate (capped, read-only); export JSON/SQL/dbt/dbt-PR
- Contract freeze; drift gates; HMAC attestation + verify endpoint

### Ops

- Health, CORS lock, prod boot secret checks, OpenAPI stub, diligence/smoke/Playwright tests, docs PDFs

## 6. UI / UX technical review

**Works:** Sunset Clay coherence; Jobs light notebook; clear primary nav; roadmap dialog aids POC demos; Chat → Jobs deep-link.

**SE will still ding:** Settings still carries marketing stubs (billing/upgrade); members admin is functional not “enterprise console”; canvas edge *evidence* must be obvious in demos; don’t oversell “AI wrote production dbt.”

| Surface                     | Status  | Demo risk                 |
|-----------------------------|---------|---------------------------|
| Login / register            | Built   | Low                       |
| Workspace / Sources / Sync  | Built   | Low if fixtures ready     |
| Join promote evidence panel | Partial | <b>High</b> — trust hinge |
| Chat → job save             | Built   | Medium (draft quality)    |
| Jobs export / attestation   | Built   | Low if GH optional        |
| Settings billing stubs      | Stub    | Medium — hide for clients |

## 7. Production-readiness sell gate

### GREEN Good enough for controlled POC

- Auth register/login, roles, invites, workspace create
- Cred encryption; prod boot refuses missing secrets / auth bypass / empty CORS
- Sync → HITL → jobs → attested export
- Scrub/drift gates; automated tests; setup + feature docs

### YELLOW Sell only with SOW constraints

- Join quality proven on eval golden-set, not yet on every customer warehouse
- Live DBX/SF = their network/tokens; fixtures ≠ production proof
- dbt PR needs GH token + repo; OpenAPI incomplete; Node+PG not full SaaS control plane
- CI workflow may need GitHub `workflow` scope to land on remote

### RED Do not claim

- SOC2 Type II / continuous compliance pack
- SCIM / directory sync
- “Full lineage / replace Collibra/Atlan”
- Unrestricted live SQL from chat
- Multi-region HA, tenant isolation audit, pen-test report, contractual SLAs

**POC success definition:** N promoted joins + 1 attested dbt (or JSON) export + precision/recall on *their* golden join pairs — not “catalog coverage %.”

## 8. Near-term tech improvements (beat the wedge)

Do these **before** chasing Atlan/Collibra parity.

| #    | Item                                                                | Why                          |
|------|---------------------------------------------------------------------|------------------------------|
| P0.1 | Join evidence UX on every edge (signals + optional SQL snippet)     | Trust hinge in demos         |
| P0.2 | Customer golden-set eval harness → POC report                       | Investor/SE proof            |
| P0.3 | One-click POC pack (migrate, fixtures, verify attestation)          | Time-to-value                |
| P0.4 | Hide billing/usage stubs outside DEV                                | Credibility                  |
| P0.5 | Audit log export + session revoke + rate limits                     | Basic enterprise hygiene     |
| P1.1 | Snowflake query-history join mining (parity with DBX)               | Cross-source wedge           |
| P1.2 | Deep dbt manifest round-trip                                        | Ship into their repo reality |
| P1.3 | Column-impact GitHub check on PR                                    | Attestation visible in CI    |
| P1.4 | Always print attestation fingerprint + approved join IDs in PR body | Non-repudiation              |

## 9. Future: compete with Atlan / Collibra / Alation / DataHub

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“Replace Atlan/Collibra this quarter” is a **category claim**. It requires becoming a *catalog + governance platform*, not only a join-shipping appliance. Build only if strategy explicitly expands — after wedge revenue.

### A. Catalog breadth (parity table stakes)

- Connectors at scale: BigQuery, Redshift, Synapse, Glue, Kafka, S3 lakes, Airflow/OpenLineage ingest
- Asset types beyond tables: dashboards, metrics, ML features, pipelines, dashboards as first-class nodes
- Search that ranks popularity/usage; duplicate dataset detection
- Business glossary + term↔table linking; stewardship workflows (certify, expire, owner queues)
- Automated multi-hop **column lineage** across SQL + dbt + BI (expensive; expected)
- Quality / profiling signals (null rates, freshness) — optional, privacy-gated

### B. Governance & collaboration (Collibra/Alation territory)

- Policy packs: PII tags, retention, access request tickets
- Workflow engine: approvals, RACI, scheduled digests (email reports)
- Integrations: Jira/ServiceNow, Slack/Teams notifications
- Domain / data-product modeling (not Draw.io stubs)
- Non-technical UX modes (steward vs DE) without diluting DE wedge

### C. AI that enterprises will trust

- Grounded answers with citation to lineage edges + owners
- Eval harnesses for hallucination; schema-only guarantees contractually encoded
- Optional “propose PR” with mandatory HITL (keep Que’s DNA)

**Do not:** market glossary + lineage depth you have not built. Competitors already lose deals on incomplete lineage — overclaiming is fatal.

## 10. Future: Big-4 audit bar multi-tenant SaaS

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Unattended multi-tenant SaaS sold into Big-4-audited enterprises needs a **control plane + evidence pack**, not just features.

### D. Identity & access (beyond invites)

- **SCIM 2.0** provisioning/deprovisioning from Okta/Entra/Google
- SSO enforced org-wide; break-glass accounts; session revoke UI; device/session inventory
- Fine-grained RBAC/ABAC; API keys with scopes; least-privilege service accounts
- Customer-managed keys (CMK) option for secrets & sample scrub salt

### **E. Security & compliance evidence**

- **SOC 2 Type II** (then ISO 27001 if EU) — policies, control testing, auditor access
- Pen-test reports (annual); vulnerability management; SBOM; dependency scanning in CI
- Immutable audit logs (who synced, promoted, exported, verified attestation) — exportable
- Data residency options; DPA/SCCs; subprocessors list
- Encryption in transit everywhere; TLS config documented; secret rotation runbooks

### **F. Multi-tenant reliability & isolation**

- Hard tenant isolation (schema-per-tenant or verified RLS) + automated isolation tests
- HA Postgres; backups/PITR; DR RPO/RTO documented and drilled
- Horizontal API; queues for sync/export; rate limits; abuse detection
- Observability: metrics, traces, alerts, status page; on-call
- Contractual SLAs (uptime, support tiers) with monitoring evidence

### **G. Enterprise delivery modes**

- Dedicated VPC / private link; optional self-hosted / hybrid gateway for warehouses
- Admin console: org settings, SSO/SCIM, audit export, feature flags
- Complete versioned OpenAPI + webhook signing

Big-4 buyers ask for **evidence**, not slideware. Budget calendar time for audit, not only engineering features.



11. Phased roadmap (do not scramble)

| Phase               | Goal                       | Build focus                                                    | Claim allowed                                                |
|---------------------|----------------------------|----------------------------------------------------------------|--------------------------------------------------------------|
| 0 — Now             | Paid design-partner POC    | Wedge reliability, evidence UX, golden-set reports, hide stubs | "Join HITL → freeze → attested PR"                           |
| 1 — Wedge scale     | Repeatable DE POCs         | SF query history, deeper dbt, GH checks, audit export          | "Cross-source join contracts for DE teams"                   |
| 2 — Platform light  | Expand without catalog lie | More connectors, BI lineage depth, better admin                | "Works beside your catalog"                                  |
| 3 — Catalog compete | Only if strategy expands   | Glossary, stewardship, column lineage at scale (\$9)           | Cautious catalog language — still not "replace this quarter" |
| 4 — Big-4 SaaS      | Unattended enterprise      | SCIM, SOC2 T2, HA/isolation, pen-test, SLAs (\$10)             | "Enterprise-ready multi-tenant"                              |

Jumping to Phase 3–4 language while shipping Phase 0 software is how diligence fails.

12. Claims you may / may not make

| Allowed now                                                                                                                                                                      | Forbidden now                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Schema-only metadata plane<br>Human-approved joins<br>Contract freeze + HMAC attestation<br>Optional dbt PR export<br>Encrypted warehouse credentials<br>Scrubbed/capped samples | Replace Atlan/Colibra/Alation this quarter<br>Full enterprise column lineage everywhere<br>SOC2 Type II certified<br>SCIM / directory sync complete<br>Unattended Big-4 multi-tenant SaaS<br>Chat = unrestricted live warehouse explorer |

13. Appendix — test & doc pointers

| Artifact                    | Path                                        |
|-----------------------------|---------------------------------------------|
| Manual POC checklist        | docs/MANUAL_TEST_PLAN.md                    |
| Testing clarity (new flows) | docs/TESTING_CLARITY.md                     |
| Feature oracles             | docs/Que-Feature-Guide.pdf                  |
| Local setup                 | docs/Que-Local-Setup-Guide.pdf              |
| API diligence               | npm run test:diligence · npm run test:smoke |
| Browser E2E                 | npm run test:e2e (UI :5174, API :8787)      |
| In-app roadmap              | Header ? after login                        |